

```
!pip uninstall -y transformers
!pip install -U transformers accelerate bitsandbytes sentencepiece
```

Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.1)
Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from transformers) (0.25.1)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.8.0)
Requirement already satisfied: tqdm>=4.60 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from accelerate) (2.11.0+cu128)
Requirement already satisfied: click>=8.4.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (8.1.8)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (3.16.1)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (2024.10.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (1.2.2)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (0.27.0)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (4.12.2)
Requirement already satisfied: setuptools>=65.5.0 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (75.1.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.4.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.1.6)
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudnn,nccl]>=12.8.1->torch>=2.0.0->accelerate) (12.8.1)
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (12.9.4)
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (9.19.0.56)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (2.28.9)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.4.5)
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.6.0)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudnn,nccl]>=12.8.1->torch>=2.0.0->accelerate) (12.8.4.1)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudnn,nccl]>=12.8.1->torch>=2.0.0->accelerate) (12.8.90)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudnn,nccl]>=12.8.1->torch>=2.0.0->accelerate) (11.3.3.83)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudnn,nccl]>=12.8.1->torch>=2.0.0->accelerate) (1.13.1.3)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudnn,nccl]>=12.8.1->torch>=2.0.0->accelerate) (12.8.90)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudnn,nccl]>=12.8.1->torch>=2.0.0->accelerate) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudnn,nccl]>=12.8.1->torch>=2.0.0->accelerate) (11.7.3.90)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudnn,nccl]>=12.8.1->torch>=2.0.0->accelerate) (12.5.8.93)
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Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudnn,nccl]>=12.8.1->torch>=2.0.0->accelerate) (12.8.93)
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Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (0.0.2)
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12.9.4->torch>=2.0.0->accelerate) (1.1.0)
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Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (2025.11.12)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (1.0.7)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (3.10)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (0.14.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer->transformers) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer->transformers) (2.18.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=2.0.0->accelerate) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch>=2.0.0->accelerate) (3.0.2)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers) (0.1.2)
Using cached transformers-5.12.0-py3-none-any.whl (11.2 MB)
Installing collected packages: transformers
Successfully installed transformers-5.12.0

```
import torch
from transformers import AutoModelForCausalLM, AutoTokenizer, BitsAndBytesConfig
import pandas as pd
import re
```

```
def extract_final_num(text):
    match = re.search(r"Final\s+Answer:\s*(-?\d+\.?\d*)", text, re.IGNORECASE)
    if match:
        return match.group(1).strip('.')
    nums = re.findall(r'-?\d+\.?\d*', text)
    if nums:
        return nums[-1].strip('.')
    return None
```

```
test_df = pd.read_csv("unified_svamp_test.csv").head(200)
```

```
import os
os.environ['LD_LIBRARY_PATH'] += ':/usr/local/cuda-13.0/lib64'
```

```
from google.colab import userdata
from huggingface_hub import login
```

```
hf_token = userdata.get('hf_token')
login(hf_token)
print("Authenticated successfully!")
```

Authenticated successfully!

```
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM, BitsAndBytesConfig
model_id = "microsoft/Phi-3-mini-4k-instruct"
bnb_config = BitsAndBytesConfig(
    load_in_4bit = True,
    bnb_4bit_quant_type = "nf4",
    bnb_4bit_compute_dtype=torch.float16
)
```

```
tokenizer = AutoTokenizer.from_pretrained(model_id, add_bos_token=False)
stop_tokens = ["<|end|>", "<|endoftext|>"]
eos_token_ids = [tokenizer.convert_tokens_to_ids(t) for t in stop_tokens if tokenizer.convert_tokens_to_ids(t) is not None]
```

```
print("Downloading the model")
model = AutoModelForCausalLM.from_pretrained(model_id, quantization_config=bnb_config, device_map="auto")
print("Success! The engine is running")
```

Downloading the model

Loading weights: 100%

195/195 [00:23<00:00, 7.07it/s]

Success! The engine is running

```
results = []
correct_count = 0
total_count = len(test_df)
```

```
print("\nStarting Baseline Evaluation-->")
for index, row in test_df.iterrows():
    question = row['question']
    ground_truth = str(row['answer']).strip()
    messages = [
        {"role": "user", "content": f"You are an expert math tutor. Question: {question}\nAnswer the question step-by-step. You"}
    ]
    prompt = tokenizer.apply_chat_template(messages, tokenize=False, add_generation_prompt=True)
    inputs = tokenizer(prompt, return_tensors="pt").to("cuda")
    outputs = model.generate(
        **inputs,
        max_new_tokens=300,
        temperature=0.1,
        do_sample=False,
        eos_token_id=eos_token_ids,
    )
    response = tokenizer.decode(outputs[0][inputs['input_ids'].shape[-1]:], skip_special_tokens=True)
    extracted_answer = extract_final_num(response)

    is_correct = False
    if extracted_answer is not None:
        try:
            if float(extracted_answer) == float(ground_truth):
                is_correct = True
                correct_count += 1
        except ValueError:
            pass

    print(f"\n--- Problem {index + 1} ---")
    print(f"Question: {question}")
    print(f"Model Output:\n{response.strip()}")
    print(f"Extracted Answer: {extracted_answer}")
    print(f"Ground Truth Answer: {ground_truth}")
    print(f"Correct: {is_correct}")
    print("-" * 30)
```

```
97 (total alligators) - 40 (hiding alligators) = 57 alligators not hiding.
```

```
Final Answer: 57.
Extracted Answer: 57
Ground Truth Answer: 57
Correct: True
-----
```

```
--- Problem 199 ---
```

Question: Shirley sold 25 cases of boxes of trefoils. How many boxes of trefoils does she need to deliver if each case has

Model Output:

Step 1: Determine the total number of boxes in one case.
Each case contains 14 boxes of trefoils.

Step 2: Calculate the total number of boxes in all the cases sold.
Shirley sold 25 cases, so we need to multiply the number of cases by the number of boxes in one case.

Total boxes = Number of cases * Number of boxes per case
Total boxes = 25 cases * 14 boxes per case

Step 3: Perform the multiplication to find the total number of boxes.
Total boxes = 25 * 14
Total boxes = 350

```
Final Answer: 350
Extracted Answer: 350
Ground Truth Answer: 350
Correct: True
-----
```

```
--- Problem 200 ---
```

Question: Jesse's room is 19 feet wide and 20 feet long. How much longer is her room than it is wide?

Model Output:

Step 1: Identify the dimensions of the room.
Jesse's room is 19 feet wide and 20 feet long.

Step 2: Calculate the difference between the length and the width.
Difference = Length - Width
Difference = 20 feet - 19 feet

Step 3: Solve the difference.
Difference = 1 foot

```
Final Answer: 1
Extracted Answer: 1
Ground Truth Answer: 1
Correct: True
-----
```

```
accuracy = (correct_count / total_count) * 100
print(f"FINAL BASELINE ACCURACY: {accuracy:.2f}% ({correct_count}/{total_count})")
```

```
FINAL BASELINE ACCURACY: 85.00% (170/200)
```